



Deuterium–argon admixture for plasma focus neutron generation

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Abstract The plasma pinch generated by plasma focus discharge features a spatially asymmetric ion acceleration due to the rapid change of the localized electromagnetic field. The neutron emission is attributed to deuteron beam interaction at the proximities of the plasma pinch and thermonuclear reaction in the pinch. Neutron is produced by more than the thermonuclear reaction, with two phases of neutron emissions clearly identified to have correlated with at least two different mechanisms. We investigated the impact of a high mass percentage of argon in the deuterium plasma pinch on the resulting neutron emission and the breakdown initiated in the electrode system. The electrical breakdown phenomenon is initiated at the backwall of the electrode system assisted by the surface current tracking of the insulator. At the same pressure with different mass ratios of the deuterium–argon mixture, the breakdown voltage is measured, and the corresponding neutron emission is analyzed. Neutron yield was measured by a calibrated indium activation counter placed at an angle from the axis of the electrodes. At a mass ratio of 50% deuterium and 50% argon, the highest average neutron yield was measured which was $(3.0 \pm 0.6) \times 10^7$ neutrons per shot, and an average breakdown voltage of 11.8 ± 1.8 kV and an average energy into the pinch of 139 ± 16 J were calculated. The pure deuterium discharge yielded an average $(4.7 \pm 0.8) \times 10^6$ neutrons per shot, while the average breakdown voltage of 12.1 ± 1.6 kV and the average energy into the pinch of 83 ± 17 J were calculated. The 50% argon admixture could increase the energy into the pinch to more than 60%, resulting in more than 5 times higher neutron yield. The ion temperature evaluated from the computation code was 1 keV, inversely proportional to the percent mass of argon. The intriguing plasma focus phenomenon is related to the implosion mechanisms and the plasma instabilities that are strongly affected by the participating ion species.

1 Introduction

In a plasma focus (PF) discharge, a well-formed plasma pinch has a density of typically 10^{18} – 10^{19} particles/cm³ and an ion temperature of around 1 keV [1, 2]. This high-density and high-temperature plasma pinch is often considered as a point source of copious radiation. The plasma focus pinch emits neutrons when deuterium or deuterium–tritium is used as the filling gas. Up to a total of 10^9 neutrons per pulse in the Filippov-type [3] and 10^{10} neutrons per pulse in the Mather-type [4] plasma focus device were reported in the early research around the 1960s. The neutron yield of megajoules PF machines was found to have an empirical relation with the n^{th} power of the pinch current where n is a factor between 4 and 5 [5]. Experimental observation of the deuterium doped with heavy gas discharges in low [6] and medium [7] energy PF revealed two distinct phases of neutron emission. The two phases of the neutrons of different spatial anisotropy and the proportions of the two varied with the percentage of doping and the mass of the dopant gas. The initial period of emission commenced during the final stage of the radial compression of the plasma column, before the formation of the hot plasma pinch column. Neutron production was primarily attributed to the radial acceleration of ions through a Fermi mechanism, where they gained energy by interacting with the incoming sheath [8]. The subsequent phase of neutron emission occurred after the initial compression, while the dense plasma column expanded as was shown in Schlieren measurements [9]. Neutron production may be correlated with the interaction between the accelerated ion beam and the hot target. Two distinct pinch modes were reported on the SPEED2 PF [10] using neon and argon-doped deuterium, named micro-pinch mode (MPM) [11] for argon doping, and stable column mode (SCM) [12] for neon doping discharges. The reported neutron yield of the SCM was consistently higher than that of the MPM. Argon doping in deuterium discharge on a 400 J PF [13] was found to have eased the initial stage of the discharge, and a decrease in the breakdown voltage was observed. Neutron yield enhancement was reported with 30% of argon doping in deuterium with 65% of the neutron yield attributed to beam–target mechanism [6], while 30%

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krypton doping in deuterium enhanced neutron yield up to 30 times [14], probably due to enhanced pinch stabilization and higher plasma density facilitated by radiative collapse.

The energy spectra of deuteron generated in a 44 kJ MAJA-PF utilizing 5% argon doping [15] ranged from 300 to 700 keV. Investigation on PF-24 with pure deuterium and deuterium–argon mixtures of different percentages, however, showed no improvement in neutron yield by the doping [16]. In the case of neon–deuterium mixture, the radius of the plasma column changed with the fraction of neon doping, while D–D fusion neutron yield decreased with an increase in the fraction of neon [17]. Adding helium, neon, and argon to deuterium discharge in a 500 J PF [18] reported no increment in neutron yield with helium doping, but a fivefold and eightfold increase in neutron yield with less than 20% mass of Ne and Ar into deuterium. Experiments conducted with Dena PF with 17.5% mass krypton addition achieved a threefold rise in neutron production [19]. The underlying mechanisms suggest that high Z gas doping could enhance current and pinching in PF discharge. The highest average neutron yield of 2.2×10^9 neutrons per shot was attained at 1.3 mbar of deuterium with about 40% mass krypton from the Sahand PF device of 90 kJ of maximum stored energy [20], and the anisotropy of neutron emission indicated that the neutrons were predominantly generated through the beam–target mechanism. For strong radiation from krypton, a lower threshold current is required for strong radiative collapse to be observed. The characteristic depletion time of the pinch energy by the radiation should be of the order of the pinch time that explained krypton doping yielded better results, and the effect of doping was significant in the large PF. The radiative collapse with krypton in a kJ PF was computed [21] by using the Lee model code and showed a minimum radius ratio of 0.0014. Validation of neutron emission could be performed with PF at several kJ of energy, as the physics is similar to a scaled-up PF of MJ [22, 23]. The filament-like structures arising from discharges of a 3 kJ PF device operated with pure deuterium were resolved by a Schlieren system and differential laser interferometry [24] and shown to be correlated with the compression, pinch stagnation, instabilities development, and disruption of the PF column. The filaments and the dynamics related to them could be an energy source for fusion reaction in the pinch. The spontaneous generation and alteration of magnetic fields during the pinch implosion, stagnation, and instabilities evolution reflected in the dynamically changing filamentary structures could result in the acceleration of charged particles.

In this work, the intriguing plasma focus phenomenon that is strongly affected by the participating ion species is investigated in different mass ratios of argon and deuterium. The participation of the high Z gas in enabling a better plasma pinch that led to higher neutron emission is determined. Simulations with the Lee model code by using the measured discharge current waveform provide information on the ion temperature during the pinch. Discharges with pure deuterium and deuterium–argon mixtures at various argon mass percentages were analyzed based on the simultaneous measurement of the current and voltage waveforms, and neutron emission.

2 Methodology

2.1 Experimental method

A Mather-type PF device is powered by a single 30 μ F capacitor charged to a maximum of 14 kV. The central anode has a vertically drilled hole with a diameter of 17 mm to a depth of 40 mm. The anode rod has a diameter of 19 mm and an effective length of 160 mm when installed into the experimental chamber, surrounded by six cathode rods at a radius of 32 mm. The cathode is made of solid copper rods, each with a length of 160 mm and a diameter of 9.5 mm. An insulator sleeve is employed to assist in the breakdown at the backwall, and it is made of Pyrex glass, 22 mm in diameter, 45 mm in effective length when installed into the experimental chamber, and a thickness of 2 mm. A resistive high-voltage probe, a Rogowski coil, and an indium activation detector are employed for measuring the discharge voltage, current, and time-integrated neutron measurements. The resistive high-voltage probe and Rogowski coil are placed at the collector base plate of the PF device. The activation counter is positioned at the z-axis of the anode at a distance of 28 cm from the tip of the anode (refer Fig. 1). The calculated voltage calibration factor for the resistive high-voltage probe and the voltage-to-current conversion factor for the Rogowski coil are 0.71 and 36 kA/V, respectively. The thin-wall indium activation counter is positioned on the z-axis of the system at a distance of 28 cm from the anode face (Fig. 1). The surface of the active area of the counter, relative to the center of the anode face and z-axis, covers approximately 0.17. The counter has been calibrated in the Agency of Nuclear Malaysia where the calculated calibration factor at a distance of 28 cm from the neutron source is 8.22×10^4 neutrons/count. The experimental setup is shown in Fig. 1. The electrical signals were recorded on a 500 MHz digital oscilloscope.

The PF chamber is filled with the operating gas, either pure deuterium or deuterium–argon mixture at the required mass ratio. The gas admixture was prepared with 10%, 30%, 50%, 60%, and 70% argon mass. The experiments were performed at a constant total pressure of 4 mbar for all the filling gases—pure deuterium and deuterium–argon mixtures. The relations between argon mass percentage ($\%m_{Ar}$), argon partial pressure (P_{Ar}), and deuterium partial pressure (P_{D2}) are as below:

Argon mass percentage :

$$\%m_{Ar} = \frac{10P_{Ar}}{10P_{Ar} + P_{D2}} \quad (1)$$

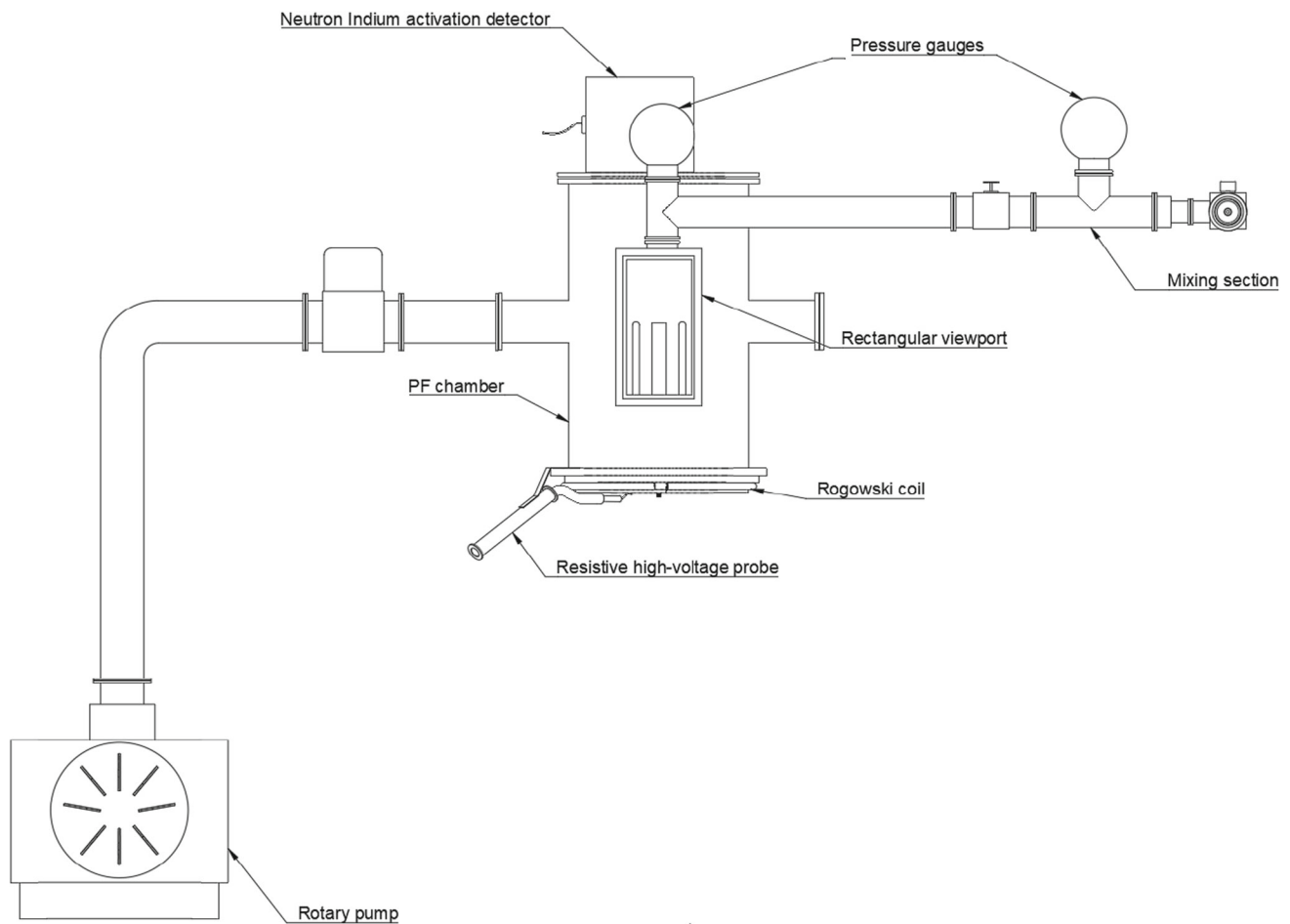


Fig. 1 The experimental setup of the plasma focus system; the PF chamber enclosing the central anode encircled by six cathode rods. The anode end has a blind hole with a depth of 40 mm

Table 1 Mass percentage and partial pressure of deuterium and argon during the mixing

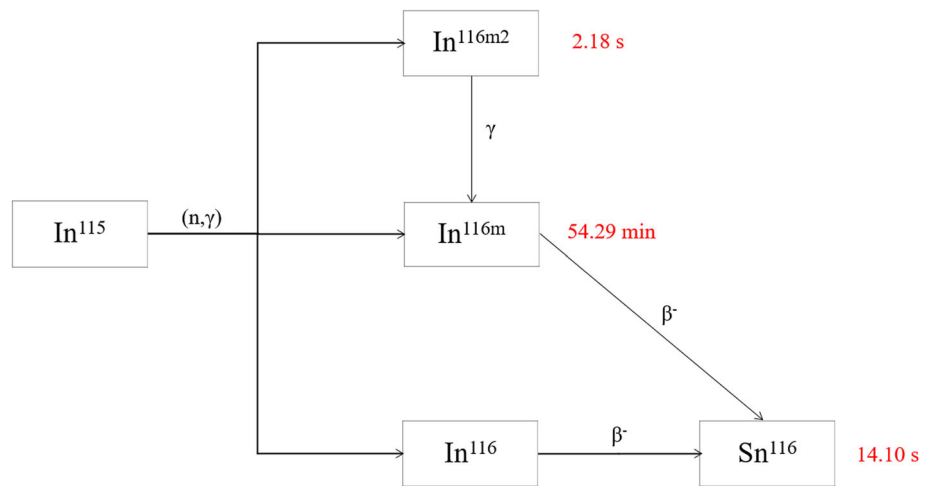
Deuterium mass percentage (%)	Argon mass percentage (%)	Deuterium partial pressure, P_{D2} (mbar)	Argon partial pressure, P_{Ar} (mbar)
90	10	900	10
70	30	700	30
50	50	500	50
40	60	400	60
30	70	300	70

Argon partial pressure :

$$P_{Ar} = \frac{\%m_{Ar} P_{D2}}{(1 - \%m_{Ar})10} \quad (2)$$

As the partial pressure and mass percentage of argon in the mixture increases, the partial pressure and mass percentage of deuterium decreases. The admixture of deuterium and argon was prepared in a small mixing chamber isolated from the main chamber by a needle valve. Table 1 shows the mass percentages of deuterium and argon and the partial pressures of the two gases during the mixing. Then, an amount of this gas mixture was flowed into the main chamber such that the amount was equivalent to 4 mbar in the main chamber. The main chamber was pumped to high vacuum after one series of experiments, which was fixed to six discharges. The main chamber was refilled to 4 mbar after around 5 min of pumping.

Fig. 2 Activation and decay scheme of In^{115} ; the beta activity of In^{116} was used in the integrated measurement, corresponding to about 25% of the branching fraction [26]



2.2 Numerical and analysis method

The plasma focus 5-phase Lee model code for gas mixture was formulated with the input parameters according to the experimental arrangement. The measured discharge current waveform was fitted in the model with computed discharge current waveform, showing the plasma pinch formation at where the current drop was measured. All equations used in the Lee model computations describing the discharge voltage, discharge current, and current sheath motion in the axial and radial phases as well as the fitting procedure can be found in the following article [25]. Two sets of current and voltage signals were analyzed, one from the experiment and one from the simulation with the Lee model code. In both pure deuterium and deuterium–argon mixture discharge, the experimentally measured waveforms showed multiple spikes in the voltage signals and drops in the current signals that were generally larger than the simulated signals.

In the consideration of energy into the pinch, the experimental data were analyzed. The product of the discharge current and voltage changed which correspond to the plasma focus gave the time dependent power of the plasma pinching. The first voltage spike was used in the computation, in the case of multiple spikes obtained, and the subsequent smaller spikes would not be included in the calculation. The phenomenon related to multiple possibilities after the main plasma focus includes secondary and usually weaker current sheaths that proceed after the main current sheath. The Lee model code does not consider the secondary current sheath formation. Therefore, all computations are based on the first plasma focus pinch formation and its compression. The computed pinch voltage, pinch current, pinch lifetime, pinch radius, energy into the pinch, and ion temperature were obtained from the discharge current fitting with the Lee model code.

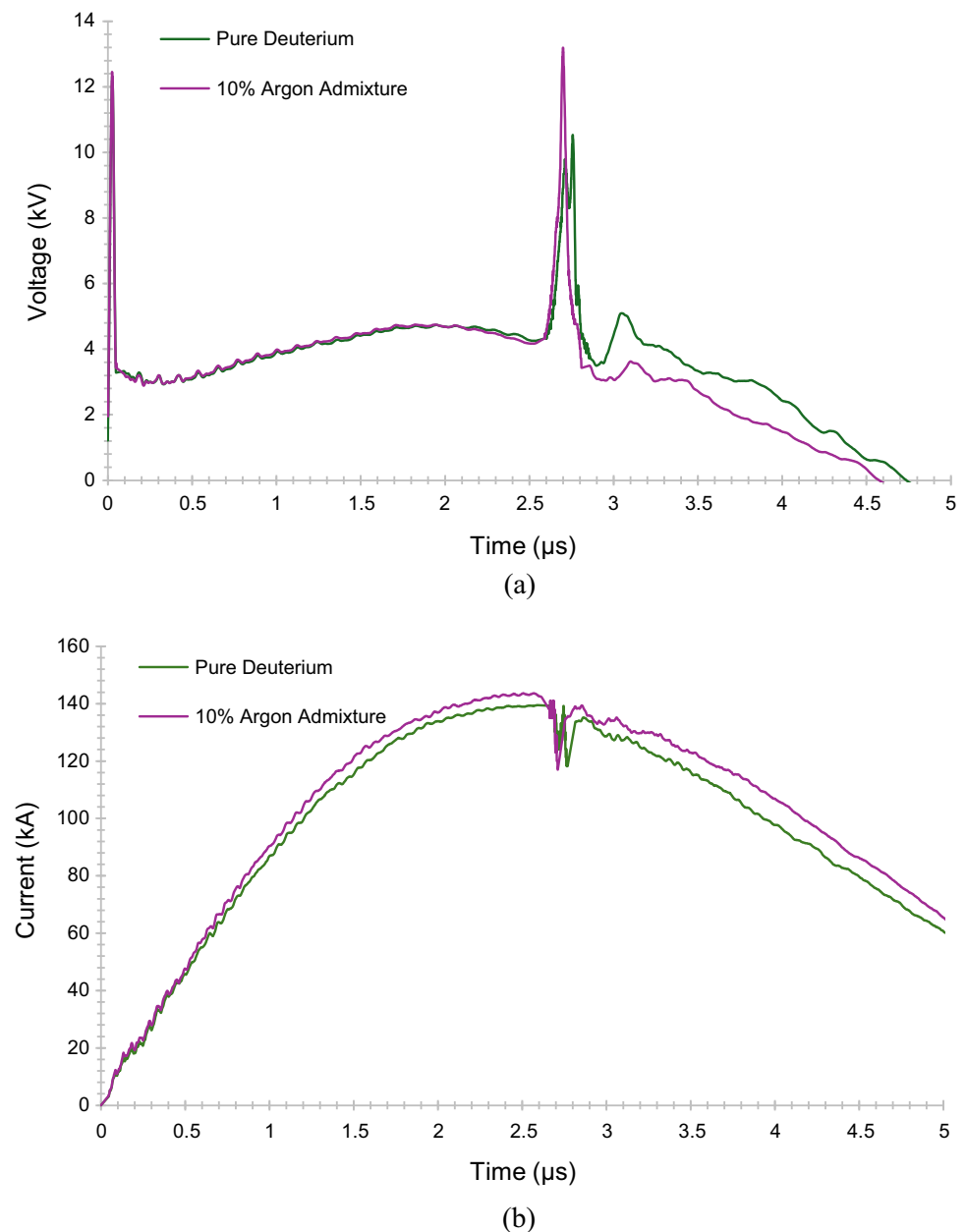
2.3 Time integrated neutron measurement method

The indium activation method was employed for the fast neutron measurement. The indium foil was purchased from goodfellow.com with a 99.999% purity percentage and a thickness of 0.5 mm, wrapped over a thin-wall GM tube. Thermalization of the fast neutron was made by paraffin wax, with the thickness determined experimentally with irradiation of neutron by an americium–beryllium neutron source. A casing of paraffin wax with a thickness of 7 cm was molded to enclose the indium activation counter. The configuration of the activation counter is like in our previous report [6]. The neutron counts were measured based on the activation of indium-115 in the branch of $\text{In}^{115}(n,\beta^-)\text{Sn}^{116}$ with the 14.1 s half-life. The activation and decay scheme of In^{116} is shown in Fig. 2.

3 Results and discussion

Argon of different percent mass was prepared, with the objective to investigate the impact on deuterium discharge and the resulting neutron emission. The experiment was conducted for each admixture for 30 shots. All shots obtained were analyzed. In the good shots, the plasma focus was observed with a high-voltage spike, and clear current dip and the neutron emission counted were high. Good and consistent shots were obtained when argon mass percentages increased to 30% or more. The plasma focus time was shifted with the addition of argon, from 2.7 to 3.3 μs . The peak values of the voltage spikes and their FWHM were analyzed correspondingly.

Fig. 3 Typical measured signals for **a** discharge voltage and **b** discharge current, plotted for both pure deuterium and 10% argon mass



3.1 Pure deuterium and 10% argon mass

In small doping, 10% in argon mass, the plasma focus was compared to pure deuterium discharge. The discharge voltage and current signals are plotted in Fig. 3a, b, respectively.

Argon-doped voltage spikes were characterized by a sharper time peak compared to pure deuterium spikes. The full width at half maximum (FWHM) of the measured voltage spikes in pure deuterium was 93 ± 10 ns, and the FWHM of the measured voltage spikes for 10% argon was only 44 ± 13 ns. The smaller FWHM was accompanied by a higher average peak voltage of 13.3 kV for 10% argon and 8.1 kV for pure deuterium: approximately 60% higher. The influence of 10% argon doping was explained using the Lee model code simulation results. The average pinch lifetime for 10% argon doping increased slightly, while the final pinch radius decreased. If the final pinch radius is given as r_p and v is the local ion speed in the pinch, then the growth of $m = 0$ instability follows $t = r_p/v$ [27]. The rapid growth in the instability would result in a higher local ion acceleration. A similar observation was reported in a deuterium–krypton plasma focus study [19]. The lifetime of plasma pinch was shortened to 40 ns at about 10% mass of krypton compared to 60 ns with pure deuterium. This might be associated with the anomalous resistance stemming from the instabilities [28]. The values of pinch lifetime for pure deuterium and 10% argon doping derived from simulation were 4.90 and 4.92 ns, and the

Table 2 The FWHM, peak voltage, pinch lifetime, and pinch radius for pure deuterium and 10% argon doping

Argon mass percentage (%)	FWHM (ns) for measured voltage spikes	Measure peak voltage (kV)	Computed pinch lifetime (ns)	Computed pinch radius (mm)
0	93	8.1	4.90	1.303
10	44	13.3	4.92	1.266

Table 3 The typical current sheath parameters computed by the Lee model code

Parameter	Argon mass percentage (%)					
	0	10	30	50	60	70
f_m	0.14	0.11	0.08	0.06	0.06	0.06
f_c	0.95	0.92	0.89	0.86	0.85	0.83
f_{mr}	0.11	0.12	0.12	0.13	0.12	0.12
f_{cr}	0.82	0.82	0.83	0.85	0.84	0.84

final pinch radii were 1.306 and 1.266 mm, respectively. The physics of interplay between different ion species in the plasma pinch is unclear, while the experimental fine-tuning of the plasma focus with the addition of heavy gas is evident, summarized in Table 2.

3.2 Neutron yield correlated with argon mass percentage in deuterium discharge

The signature of plasma focus in the discharge voltage peak measured experimentally was compared to the simulated data obtained using the Lee model code. The simulated data were not taken with assumed parameters but each through careful fitting of the measured current waveform to ensure the correct plasma dynamic has been considered. The typical values of the f_m , f_c , f_{mr} , and f_{cr} parameters obtained from the simulations for discharges in pure deuterium and the deuterium–argon mixtures are summarized in Table 3.

The plots in Fig. 4 give the comparison between the experimental data and the computed average values of the pinch voltage, which corresponds to the voltage spike during the formation of plasma pinch. The error bars were obtained using the standard deviation of the measured and computed values of the pinch voltage. The standard deviation was chosen as a measure of the shot-to-shot variation so that the effect of doping to it can be inferred. The computed pinch voltage increased with increasing argon mass, from 44 to 46 kV. The experiments measured a much lower pinch voltage, also increasing with increasing argon mass, with some shots having higher voltage spikes. The average pinch voltage increased from 0 to 70% argon mass. This was correlated with tighter pinches and suggests a smaller final plasma column radius (increase in plasma compression). The magnetic pressure compressing the pinch is directly related to the current during the plasma focus. The pinch current, I_{Pinch} , for the different gas mixtures was computed folded in the factor f_{cr} . It is the magnitude of current at the maximum compression of the plasma pinch. The I_{Pinch} increased with an increasing argon mass percentage (Fig. 5) from 138 kA in pure deuterium to about 144 kA for 60% and 70% argon mass. The error bars were again obtained using the standard deviation formula. The current sheath dynamics has the important role in matching the characteristic time of the plasma focus pinch. The pinch time corresponds to the moment of maximum compression of the plasma focus column, which is the peak time of the pinch voltage spike. The pinch time is superimposed in Fig. 5. It is shown that a matched condition was met in 50% argon mass. The matched condition is approximately fulfilled when the radial compression of the plasma sheath begins (a sharp drop in current) just after the discharge current has reached the maximum possible value enabled by the capacitor bank—full utilization of the input energy [29]. Ideally, if the current waveform could be engineered to a constant peak, it would enable good consistency and reproducibility of the plasma focus. Practically, the plasma focus devices sometime behave with better consistency and reproducibility around the matched criterion, observed with less shot-to-shot variation. The results showed that at 50% argon mass, the best operating condition was matched with minimum variation (small standard variation or error bar). This has evidenced the important role of heavy ions in the deuterium discharge despite the system parameters remain unchanged.

The neutron yield recorded by the time-integrated method using the indium activation counter was analyzed and is presented in Fig. 6. The registered total neutron emission yield for pure deuterium was $(4.7 \pm 0.8) \times 10^6$ neutrons per shot. The standard deviation of the measured values was taken as the error bar. Standard deviation was chosen as to infer the effect of doping on the shot-to-shot variation. The neutron yield obtained follows the scaling law, $n \propto E^2$ (this E refers to the total energy stored in the capacitor bank) [30–33]. The presented neutron yields in Fig. 6 increase with argon mass up to about 50% of doping. The presented data are average neutron yields, and the Y-error bars represent the standard deviation of the shot-to-shot variation. The neutron yield has been most consistent in pure deuterium at about $(4.7 \pm 0.8) \times 10^6$ and at a high percentage of argon with 70% argon mass about $(1.5 \pm 0.1) \times 10^7$ neutrons per shot. This suggests a better reproducibility for neutron production when a single dominant gas type is used. The argon mass ratio for the best performance that gave high neutron yield was found to be in the range of 30–60%. Changes in the uncertainties of the average total neutron emission yield are caused by changes in discharge and plasma parameters between successive shots, as well as competition between different mechanisms affecting neutron production. The highest neutron yield

Fig. 4 Average pinch voltage measured and computed by the Lee model code

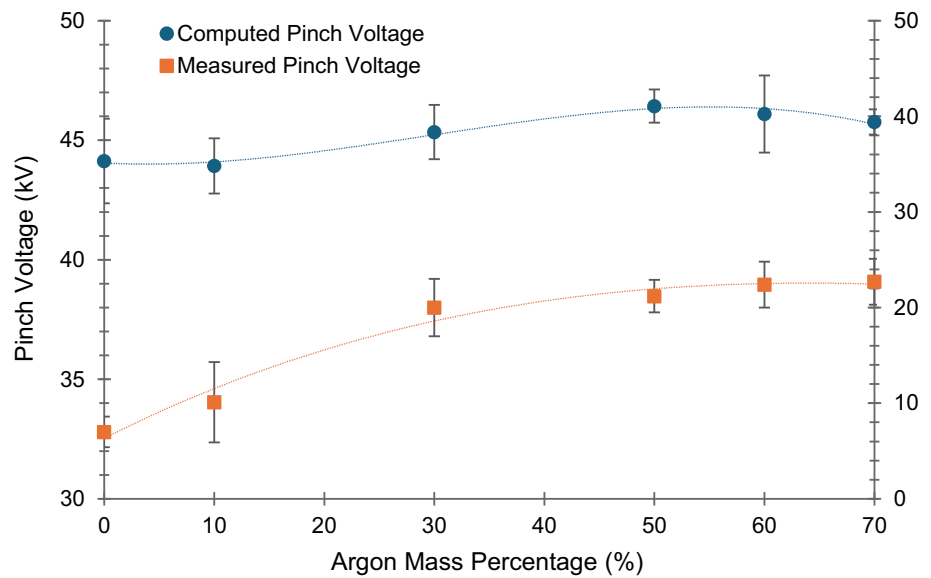
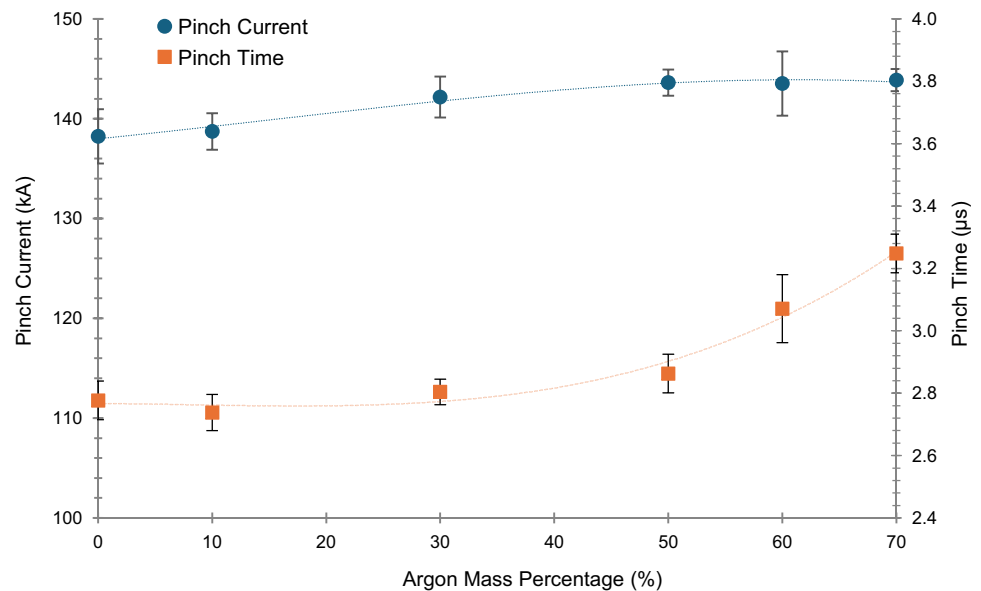


Fig. 5 Average computed pinch current (I_{Pinch}) and pinch time for increasing argon mass percentage



recorded was 3.9×10^7 with 50% argon mass, while the average yield is $(3.0 \pm 0.6) \times 10^7$ neutrons per shot. The average neutron yield is sixfold higher compared to pure deuterium discharge. The shot-to-shot variation was $\pm 20\%$ from the average value.

The pinch current in Fig. 5, average computed pinch current (I_{Pinch}) and pinch time for increasing argon mass percentage, shows 138 kA for pure deuterium and about 144 kA for 50% argon mass percentage. During discharges with pure deuterium, our plasma focus system operated according to the following relation on average: $Y \sim I_{\text{Pinch}}^{3.1}$. While during discharges with a mixture of 50% Ar and 50% D₂, the system operated significantly more efficiently and in accordance with the following relation on average: $Y \sim I_{\text{Pinch}}^{3.5}$.

The energy into the pinch evaluated for 50% argon mass was enhanced more than 60% relative to the pure deuterium. Energy into the pinch is calculated by integrating the pinch region of the discharge voltage waveform, which corresponds to the energy consumed for the compression of the plasma. It is different than the discharge energy where discharge energy is the energy transferred to the system from the capacitor bank, whereas energy into the pinch is the fraction of the discharge energy utilized for the compression of the plasma pinch. The average values of the energy into the pinch are presented in Fig. 7, average energy into the pinch calculated based on data from experimental plasma focus discharges and from simulated plasma focus discharges using the Lee model code. The computed energy into the pinch based on the Lee model code was also presented for comparison in Fig. 7, average energy into the pinch calculated based on data from experimental plasma focus discharges and from simulated plasma focus discharges using the Lee model code. It tends to linearly increase with the argon mass percentage. According to the computation, the energy into the pinch for 50% argon mass could be 95 J based on the computed results in the Lee model. The probable significantly higher energy into the

Fig. 6 Average measured total neutron emission yield for deuterium and deuterium–argon plasma focus

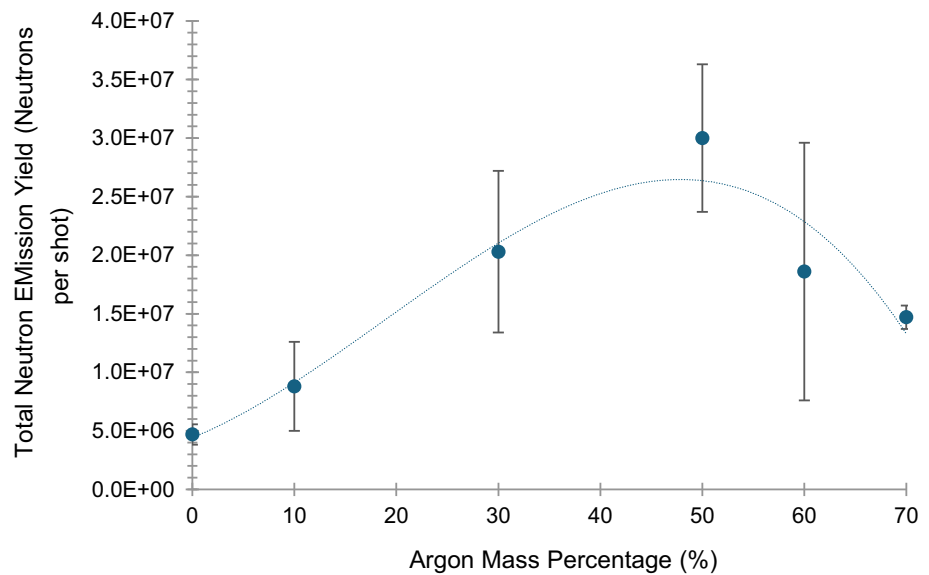
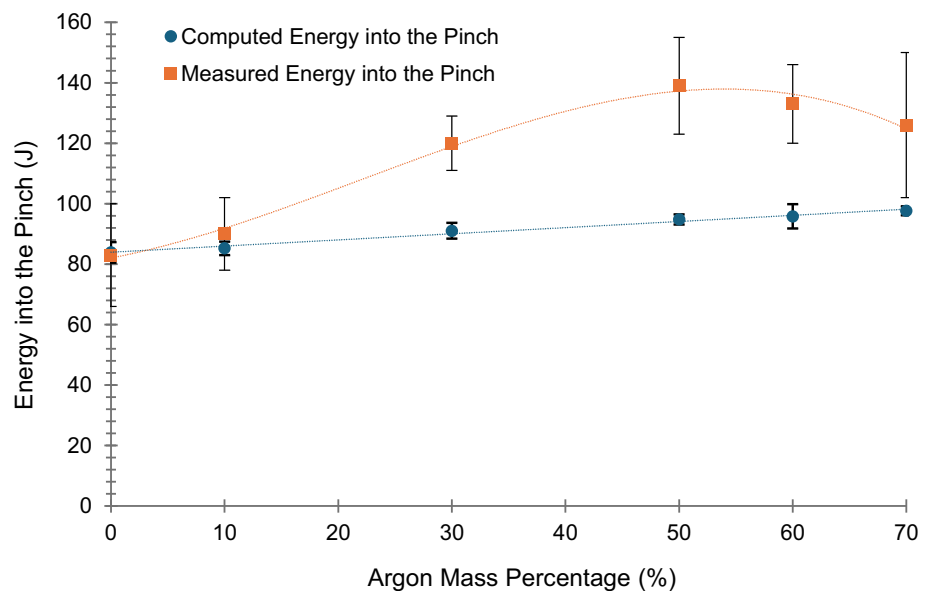


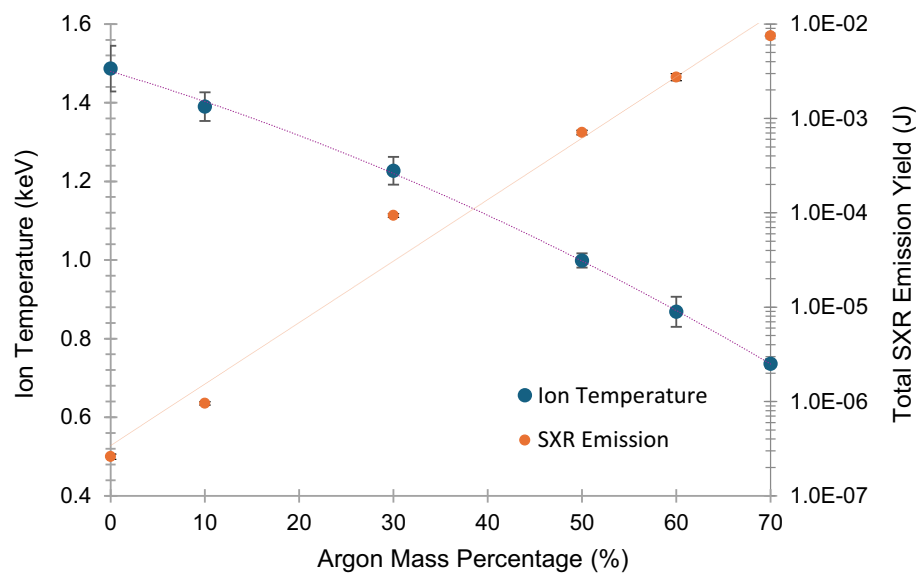
Fig. 7 Average energy into the pinch calculated based on data from experimental plasma focus discharges and from simulated plasma focus discharges using the Lee model code



pinch which results from the addition of argon ions to the plasma focus apparently has not been properly included in the computation code. But the experiments clearly suggest this higher energy based on the discharge current and voltage measurements. Considering the pinch time of less than 10 ns in the dimension of sub-mm, suitable diagnostics techniques to resolve the mechanisms have to identify the different between deuterium and argon from their emissions and masses. The energy into the pinch for each experimental data was measured from the voltage spike and current dip. The measured average energy into the pinch for pure deuterium discharge was (83 ± 17) J, compared to the measured average energy into the pinch for 50% argon mass of (139 ± 16) J. A lower energy into the pinch was observed when the argon percentage was 60% and 70%. The lower energy into the pinch in these admixtures is probably due to high radiation lost through X-ray emission over the gain in energy into the pinch through better compression. The error bars in Fig. 7, average energy into the pinch calculated based on data from experimental plasma focus discharges and from simulated plasma focus discharges using the Lee model code, correspond to the standard deviation of the measured or computed datasets.

The Lee model code was utilized to compute the ion temperature and the soft X-ray (SXR) emission by fitting the measured discharge current into the computed discharge current. The average ion temperature and SXR emission for the different filling gases are shown in Fig. 8, average computed ion plasma temperature and average computed total SXR emission yield from the pinch, simulated using the Lee model code. The X-ray emission through bremsstrahlung and line emission are prominent in inert gases like argon. The effect of energy loss is explained by the radiative cooling effect [14, 21, 34–36]. The increase in SXR emissions with increasing argon mass percentage demonstrates the increasing radiative cooling effect contributed by the argon admixture ions.

Fig. 8 Average computed ion plasma temperature and average computed total SXR emission yield from the pinch, simulated using the Lee model code



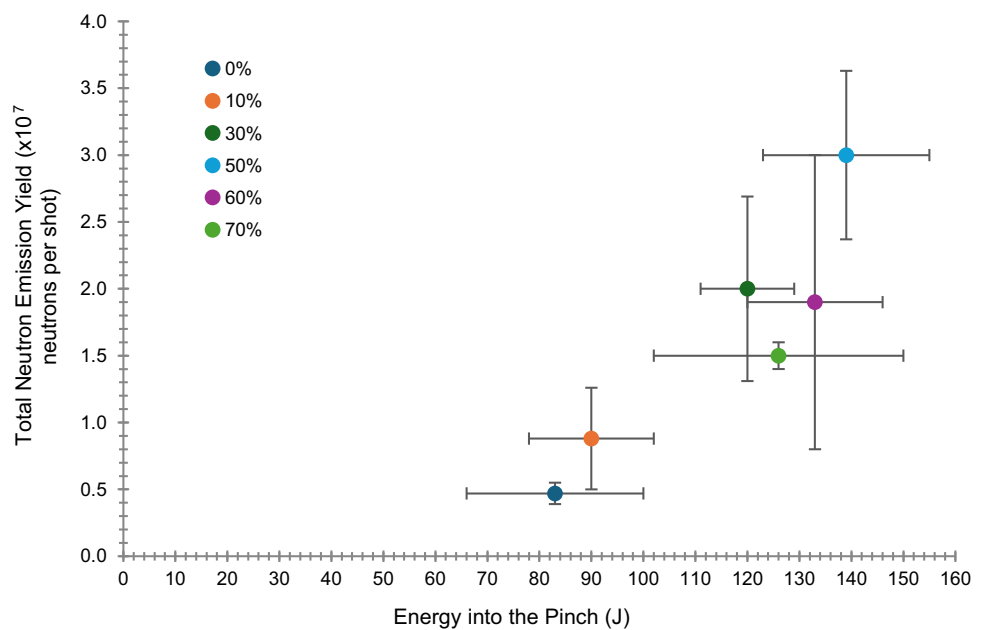
According to the obtained results, we believe that the threshold of radiation compression contributing to better plasma focus in the admixture of argon in deuterium discharge lies between 50 and 60% of argon mass. The lack of further significant plasma pinch compression at above 50% argon mass is probably related to the matching condition [29] mentioned in the earlier part of this section. Energy into the pinch ceased in higher argon mass. Similar limitations are observed also in other PF experiments [37]. The interplay between ions in the plasma focus and the accelerated ions in gyration [38] in the pinch and its vicinity due to the local magnetic field structure results in the neutron emission.

The average neutron yields are plotted together with the average energy into the pinch in Fig. 9, average measured total neutron emission yield vs average measured energy into the pinch for plasma focus discharges in pure deuterium and deuterium–argon gas mixtures, with an argon percentage ranging from 10 to 70% by mass. The highest neutron yield measured at $(3.0 \pm 0.6) \times 10^7$ neutrons per shot at argon mass of 50% corresponds to the relatively highest energy into the pinch, measured at (139 ± 16) J. The two distinct mechanisms of neutron production arising from a plasma focus are of thermonuclear origin and the beam–target reaction. The argon ions in deuterium plasma probably enhanced the radiation emission and cooling effect, leading to enhanced pinch compression and resulting in a smaller final pinch radius. This enhanced compression probably contributed to the observed pinch voltage. This could also contribute to higher acceleration voltage for the deuterons at the onset of instabilities. However, the rapid plasma focus action, particularly at the high percentage of argon mass at 60% and 70%, was observed with greater radiation cooling effect, less energy into the pinch, compared to the lower argon mass. In the hot dense plasma pinch and the onset of instabilities [39], the energetic ions [40] were responsible for the beam–target neutrons. A plasma focus would require a far higher discharge current to achieve proper pinch temperature for thermonuclear fusion becoming significant. The observed neutron emission most likely comes mainly from the interaction of high-energy tail ions with ions confined in the pinch.

4 Summary and conclusions

Measurement of the plasma focus discharge voltage, discharge current, and neutron yield for pure deuterium and deuterium–argon mixture was carried out to give a direct indication of the performance of the small plasma focus in generating neutrons. The neutron yield obtained by the time-integrated measurement was the total count from both the thermonuclear origin and non-thermonuclear mechanisms. The measured current waveforms of all the shots were carefully fitted with the Lee Mode Code and determined the respective current shielding and mass shielding factors; that the simulated data were used for discussion together with the experimental observation. The pinch current I_{Pinch} increases when argon was added into the deuterium discharge resulting in the formation of a better pinch and peak at about 50% of argon mass. The pinch voltage measured and simulated both showed increased values for higher argon mass. A significant neutron yield enhancement of 131% was measured in deuterium–argon admixture when argon mass was only 10%. Therefore, a neutron generator with plasma focus can be better with the admixture than pure deuterium under certain conditions that is commonly believed to be the only required fuel for neutron production. The interplay between argon and deuterium ions in the plasma pinch is complicated and uncertain despite the experiment's evidence of the impact in enhancing neutron production. The ion temperature of the plasma pinch computed using the Lee model code showed a decreasing trend with increasing argon mass percentage. The SXR emissions increased with increasing argon mass percentage indicating higher radiative cooling effect. This energy lost in radiations is competing with the energy into the pinch benefited by argon doping. The energy

Fig. 9 Average measured total neutron emission yield vs average measured energy into the pinch for plasma focus discharges in pure deuterium and deuterium–argon gas mixtures, with an argon percentage ranging from 10 to 70% by mass



efficiency, which is the ratio of the measured energy into the pinch to the total discharge energy (2.7 kJ), was about 5% compared to 3% for pure deuterium discharge. The pure deuterium at approximately 4.0 mbar proved to be an optimum condition for neutron production at the fixed set of parameters used here.

The high neutron yield, which is directly correlated with the performance of the plasma focus, peaked at 50% argon mass with $(3.0 \pm 0.6) \times 10^7$ neutrons per shot. This is about sixfold the average yield of pure deuterium with $(4.7 \pm 0.8) \times 10^6$ neutrons per shot. The 50% deuterium–argon mixture discharge, which consists of a mass ratio of half deuterium and half argon, has resulted in the best reproducibility of good pinches and high neutron yield. Under these conditions, the increased energy into the pinch was most effectively used for thermonuclear and beam–target fusion and neutron production. Particularly, the energetic deuteron beams were probably enhanced by the influence of the strong magnetic energy on the pinch and the onset of instabilities. A balanced and matched condition has been promoted by an optimally mixed argon and deuterium as the working gas for plasma focus discharge.

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Data availability The manuscript has associated data in a data repository.

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